

Ghost Fader

USB-MIDI-controlled volume and pan for two balanced 1/4" line channels. Bus powered, no external supply.
Sheets: 1 input, 2 matrix, 3 output, 4 control, 5 power. Nets between sheets are global labels; rails are power symbols.

Balanced inputs



File: input.kicad_sch

2x2 gain matrix and summers



File: matrix.kicad_sch

Balanced outputs



File: output.kicad_sch

Control



File: control.kicad_sch

Power



File: power.kicad_sch

Signal flow: J1/J2 -> U1/U2 THAT1246 (-6 dB) -> C3-C6 -> U3/U4 PGA2310 2x2 matrix -> R2-R9 + U5 OPA2134 summers (x2) -> U6/U7 THAT1646 (+6 dB) -> J3/J4
Control: XIAO RP2040 USB MIDI -> SPI (SCLK, SDI, /CS) + /MUTE -> U3 -> U4. Plug sense from J1/J2 ring-normal contacts.
Power: XIAO 5V pin (USB 5 V) -> FB1 -> PS1 TBA 2-0522 -> +/-12 V. Same 5 V feeds PGA logic.

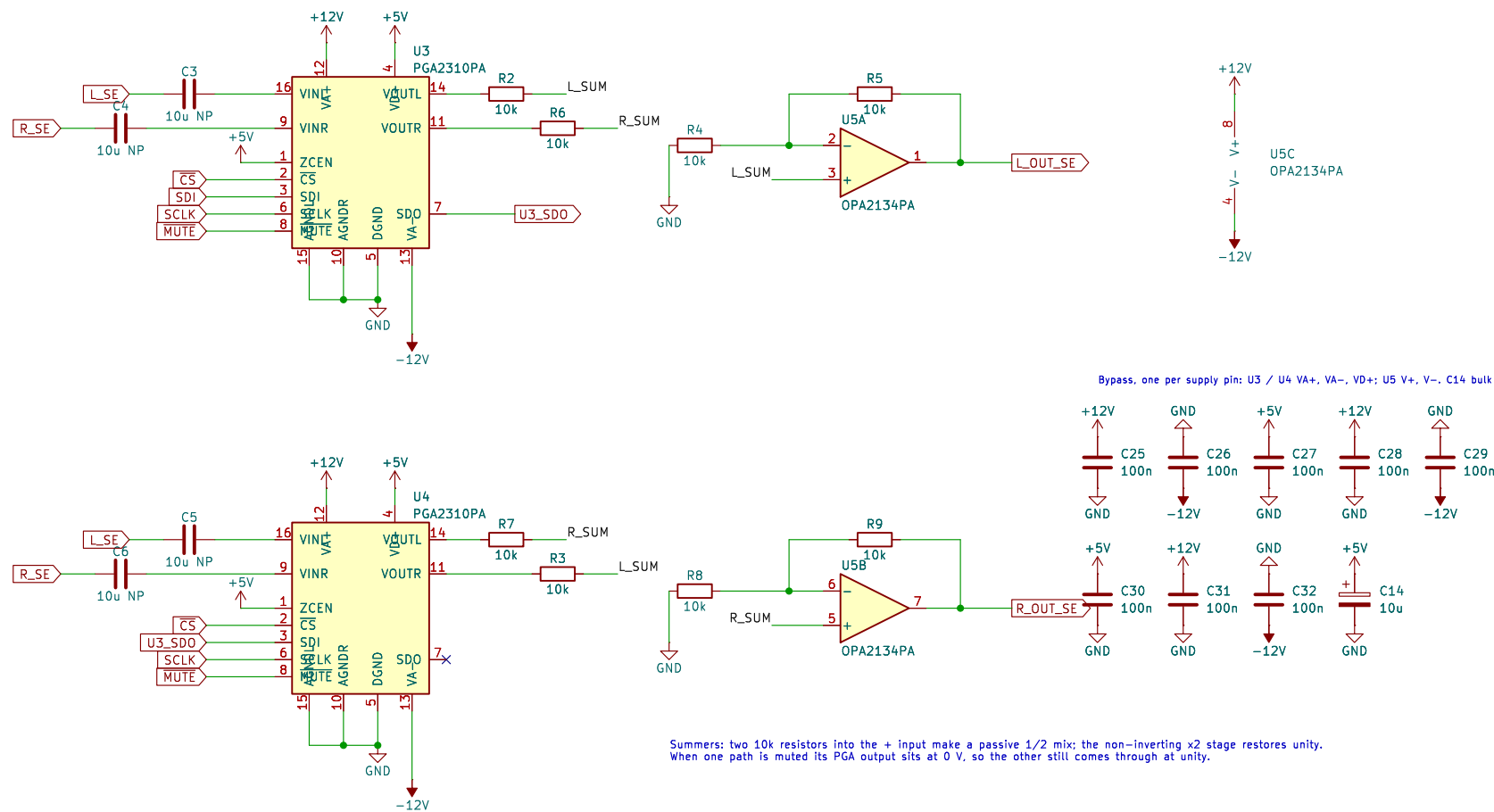
<https://chris-albert.github.io/ghost-fader/>
USB-MIDI volume and pan for two balanced line channels
Ghost Fader

Sheet: /
File: ghost-fader.kicad_sch

Title: Ghost Fader – Rev C

Size: A4	Date: 2026-09-08	Rev: C
KiCad E.D.A. 9.0.6	Id: 1/6	

Size: A3	Date: 2026-09-08	Rev: C
KiCad E.D.A. 9.0.6		Id: 2/6



U3 = direct paths (L->L on its left channel, R->R on its right). U4 = cross paths (L->R on its left channel, R->L on its right).
SPI chain: XIAO MOSI (D10) -> U3 SDI, U3 SDO -> U4 SDI. 32 clocks per update: U4 word first, then U3; each word is right byte then left byte, MSB first.
ZCEN high: gain changes wait for a zero crossing. Code 192 = 0 dB, 0 = mute. Never route SDO (5 V logic) back to the XIAO.

<https://chris-albert.github.io/ghost-fader/>
PGA2310 pair, OPA2134 summers

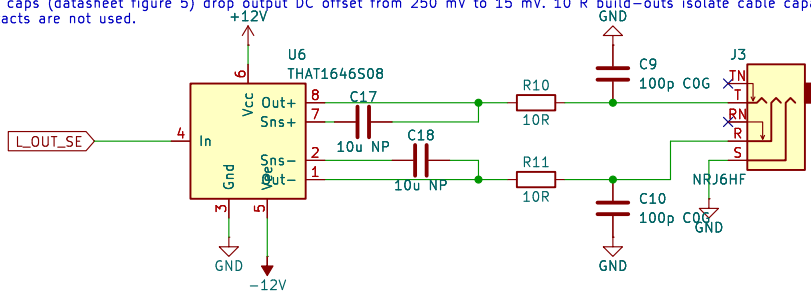
Ghost Fader

Sheet: /2x2 gain matrix and summers/
File: matrix.kicad_sch

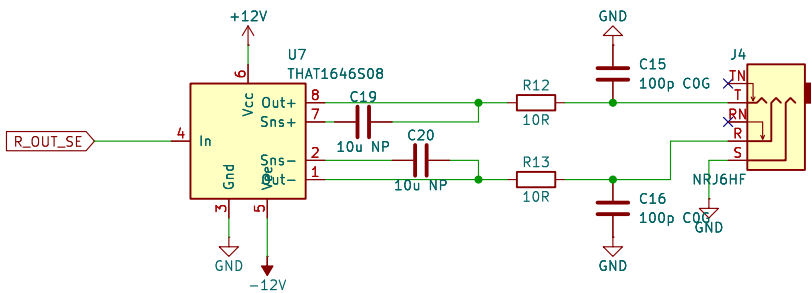
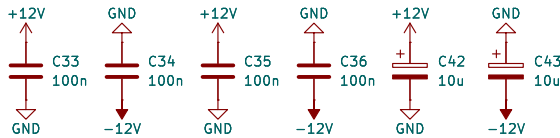
Title: Ghost Fader – Sheet 2: 2x2 gain matrix and summers

Size: A3	Date: 2026-09-08	Rev: C
KiCad E.D.A. 9.0.6		Id: 3/6

THAT 1646 = +6 dB cross-coupled driver. A TS plug on the output shorts Out- to ground; OutSmarts keeps the full signal on the tip.
10 uF non-polar sense caps (datasheet figure 5) drop output DC offset from 250 mV to 15 mV. 10 R build-outs isolate cable capacitance.
Output jack switch contacts are not used.



Bypass, one per supply pin, within 5 mm of U6 / U7. C42, C43 bulk for the output side.



<https://chris-albert.github.io/ghost-fader/>
THAT 1646 cross-coupled drivers

Ghost Fader

Sheet: /Balanced outputs/
File: output.kicad_sch

Title: Ghost Fader – Sheet 3: Balanced outputs

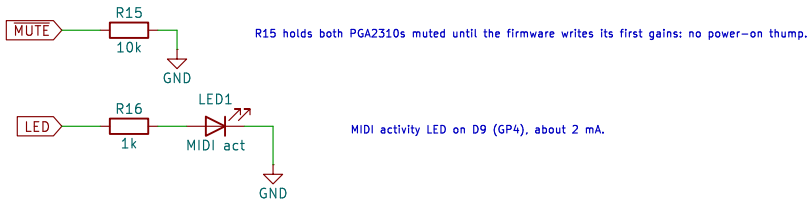
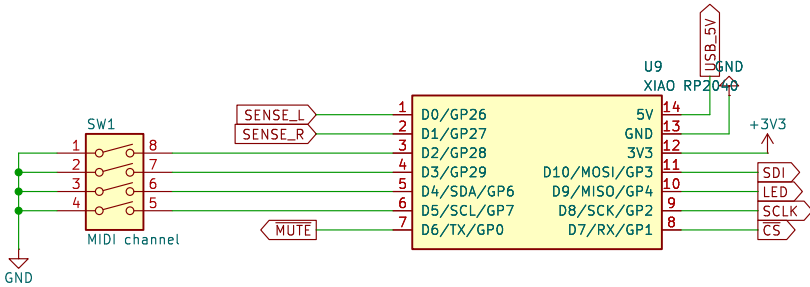
Size: A3
KiCad E.D.A. 9.0.6

Date: 2026-09-08

Rev: C

Id: 4/6

U9 is a Seeed XIAO RP2040 on header pins. Pin numbers are Seeed's: 1–7 down the left row from the USB end, 8–14 up the right row. Its USB–C is the only connector: class-compliant MIDI in (TinyUSB), 5 V in. The 5V pin is USB VBUS straight through, so it feeds the board. SPI0: D8 (GP2) SCK, D10 (GP3) MOSI, D7 (GP1) /CS, D6 (GP0) /MUTE, D9 (GP4, the MISO pin) drives the LED: bit-bang the 32-bit frame or leave SPI RX unassigned. PGA SDO is 5 V logic and must not come back. DIP switch on D2–D5 (GP28, GP29, GP6, GP7), closed = 0, internal pull-ups on. Plug sense on D0, D1 (GP26, GP27); internal pull-ups off.



<https://chris-albert.github.io/ghost-fader/>
XIAO RP2040, USB MIDI, plug sense, SPI

Ghost Fader

Sheet: /Control/
File: control.kicad_sch

Title: Ghost Fader – Sheet 4: Control

Size: A3 Date: 2026-09-08

Rev: C

KiCad E.D.A. 9.0.6

Id: 5/6

